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RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline recommends regional pooling in multi-regional clinical trials (MRCTs) based on similarity of effect modifier (EM) distributions, but provides no quantitative methodology. We propose the normalized Area Between Cumulative Distributions (nABCD), a planning-stage dissimilarity index defined as the Wasserstein-1 distance divided by twice the pooled interquartile range. Unlike the standardized mean difference (SMD), nABCD captures differences in variance, shape, and skewness—distributional features that may drive treatment effect heterogeneity through non-linear effect modification. Because nABCD requires only baseline EM distributions, it can be computed from any representative data source, including prior trials, disease registries, and real-world evidence databases. When prior evidence on conditional average treatment effect (CATE) sensitivity is available, clinical calibration translates nABCD into maximum potential regional treatment effect differences ($\Delta_{\max}$) with bootstrap confidence intervals. Simulation studies across eight scenarios demonstrated satisfactory bias and coverage of the percentile bootstrap estimator at sample sizes typical in MRCT subgroups. Application to two publicly available stroke trial datasets illustrates both common planning scenarios: Case A (EM unknown; IST-1, 31 countries) using sensitivity analysis, and Case B (EM identified; IST-3, 8 countries) using clinical calibration. The IST-3 results demonstrate that similar distributional magnitudes can yield markedly different clinical conclusions: NIHSS (nABCD = 0.240, $\Delta_{\max}$ = 5.02 percentage points) versus age (nABCD = 0.285, $\Delta_{\max}$ = 0.47 percentage points), underscoring that clinical calibration—not nABCD magnitude alone—should guide pooling decisions. The nABCD framework fills a methodological gap in ICH E17 implementation, supporting evidence-based pooling decisions from any representative population data source.

KEYWORDS

multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.[1,2] This approach offers substantial benefits: accelerated timelines, broader generalizability, and earlier access to new therapies for patients worldwide. The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs, with a central assumption that treatment effects are generalizable across the target population.[3]

A key strategy for enhancing the ability to assess regional consistency in treatment outcomes is the regional pooling approach, wherein regions with similar patient characteristics are grouped for analysis.[3] The ICH E17 guideline explicitly recommends that pooling decisions be based on the similarity of effect modifier (EM) distributions:

Abbreviations: ABCD, area between cumulative distributions; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; EM, effect modifier; EU, European Union; ICH, International Council for Harmonisation; IPD, individual patient data; IQR, interquartile range; IST-3, Third International Stroke Trial; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; nABCD, normalized area between cumulative distributions; NIHSS, National Institutes of Health Stroke Scale; NMPA, National Medical Products Administration; OHS, Oxford Handicap Scale; SMD, standardized mean difference; US, United States.

"Regions may be pooled for randomisation and/or analysis if subjects are thought to be *similar enough* with respect to intrinsic and/or extrinsic factors relevant to the disease and/or drug under study." (ICH E17, Section 2.2.5)

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. For example, if younger patients respond better to treatment than older patients, age is an effect modifier. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why EM distributional similarity is critical to the validity of regional pooling.

Despite the regulatory importance of EM distributional similarity, current practice lacks a standardized quantitative methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Recent regulatory guidance has highlighted this gap. Song et al., writing from the China NMPA perspective on ICH E17 implementation, note the challenge of operationalizing pooling criteria without quantitative tools.[4] Long et al. further discuss basic considerations for consistency evaluation under ICH E17.[5]

Current approaches to assessing distributional similarity have significant limitations (Table 1). The standardized mean difference, while widely used for baseline covariate comparisons,[6] fundamentally cannot detect differences in variance or distributional shape—precisely the types of differences that may drive treatment effect heterogeneity through effect modification.

TABLE 1 Limitations of current approaches to distributional similarity assessment.

Method	Limitation
Visual inspection	Subjective, not reproducible
Standardized mean difference (SMD)	Captures only location, ignores scale and shape
Kolmogorov–Smirnov statistic	No interpretable scale for decision-making

This paper addresses the methodological gap by proposing the *normalized Area Between Cumulative Distributions* (nABCD), a dissimilarity index for comparing EM distributions across regions, together with a clinical calibration framework that translates distributional differences into potential treatment effect heterogeneity on the outcome scale. Our specific research question is:

How can we estimate distributional similarity between regions in a scale-free manner, and translate that estimate into clinically interpretable information about potential treatment effect heterogeneity?

The emphasis is on *estimation and clinical interpretation*, not hypothesis testing. We seek to provide regulatory scientists with quantitative tools that inform deliberation, not with binary accept/reject rules. This design philosophy reflects the ICH E17 principle that pooling decisions require contextual judgment rather than mechanical application of thresholds. Importantly, because nABCD operates on baseline EM distributions rather than treatment outcomes, it functions as a *planning-stage* tool: the required distributional data can be sourced from prior clinical trials, disease registries, electronic health records, or other real-world evidence (RWE) databases—any data source that provides representative samples of the target population in each candidate region.

The nABCD index quantifies the total area between two cumulative distribution functions, normalized by the pooled interquartile range to achieve scale-free interpretation. The framework offers four contributions:

1. *Full distributional comparison*: The Wasserstein-1 distance captures differences in location, scale, and shape simultaneously.[7]
2. *Scale-free estimation*: Normalization by IQR yields a unitless measure, enabling comparisons across EMs measured on different scales. Bootstrap confidence intervals quantify estimation uncertainty.
3. *Clinical calibration*: Through the heterogeneity bound (Proposition 2), nABCD estimates can be translated into potential treatment effect differences ($\Delta_{\max}$) specific to each EM's clinical context, expressed on the primary outcome scale.[8]
4. *Sensitivity analysis*: Because the calibration depends on the CATE sensitivity parameter L, which may be uncertain, the framework naturally accommodates sensitivity analysis over L, providing a richer evidence base for decision-making than a single binary test.

We focus on continuous effect modifiers, for which the first Wasserstein distance provides a natural and theoretically grounded measure of distributional dissimilarity. Categorical or mixed-type effect modifiers (e.g., genotype, race, prior treatment history) require alternative distance measures and are beyond the scope of this work.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework. Section 3 describes a comprehensive simulation study. Section 4 demonstrates the framework using publicly available individual patient data from two international stroke trials as representative data sources, illustrating both EM-unknown and EM-identified planning scenarios. Section 5 discusses implications, limitations, and future directions.

2 | METHODS

2.1 | The nABCD Dissimilarity Index

An effect modifier (EM) is a baseline patient characteristic for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the EM in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

To formalize the relationship between distributional differences and treatment effect heterogeneity, let the CATE function be bounded with Lipschitz constant L. The difference in regional average treatment effects can be bounded by:

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2), \tag{2}$$

where $W_1(F_1, F_2)$ denotes the Wasserstein-1 distance between EM distributions in regions 1 and 2.

The Wasserstein-1 distance (also known as the Earth Mover's Distance) between two cumulative distribution functions F and G is defined as:[9]

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)|\, dx. \tag{3}$$

Geometrically, this equals the total area between the two CDFs (Figure 1). Unlike the standardized mean difference, the Wasserstein distance responds to changes in variance, skewness, and other distributional features.[7] The choice of W_1 over alternative distance measures is driven by the Kantorovich–Rubinstein duality theorem,[9] which provides the theoretical bridge between distributional distance and treatment effect heterogeneity. The connection proceeds in three steps:

1. *W_1 as CDF area*: In one dimension, $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx$ (equation 3), the total area between two CDFs.
2. *Kantorovich–Rubinstein duality*: $W_1(F_1, F_2) = \sup_{\|f\|_{\mathrm{Lip}} \leq 1} |\int f\, dF_1 - \int f\, dF_2|$, meaning W_1 equals the maximum difference in expectations over all 1-Lipschitz functions.[9]
3. *CATE application*: Since any CATE function with Lipschitz constant L satisfies τ/L being 1-Lipschitz, the duality immediately yields $|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2)$, which is equation (2).

This duality is specific to W_1. The W_2 distance, while admitting closed-form expressions for Gaussian families, does not possess this dual characterization via Lipschitz functions and therefore cannot provide the heterogeneity bound that is central to our clinical calibration framework. Similarly, information-theoretic divergences such as Kullback–Leibler (KL) divergence have structural limitations: KL divergence is asymmetric ($D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$), inconsistent with the symmetric nature of regional pooling decisions; it can diverge to infinity when supports do not overlap, a common occurrence with empirical distributions; and most critically, it lacks the Lipschitz-bound property—KL divergence has no analogous result linking it directly to treatment effect heterogeneity for Lipschitz-continuous CATE functions.[10] The Wasserstein distance itself is a true metric satisfying the triangle inequality and metrizing weak convergence.[9] The nABCD normalization by the pair-dependent pooled IQR means that nABCD does not inherit the triangle inequality—a counterexample exists for scale-differing distributions—but nABCD retains non-negativity, symmetry, and identity of indiscernibles, making it a well-defined dissimilarity index. Importantly, the Kantorovich–Rubinstein duality that underpins our clinical calibration framework operates on W_1, not on nABCD, so the theoretical bridge between distributional distance and treatment effect heterogeneity remains intact.

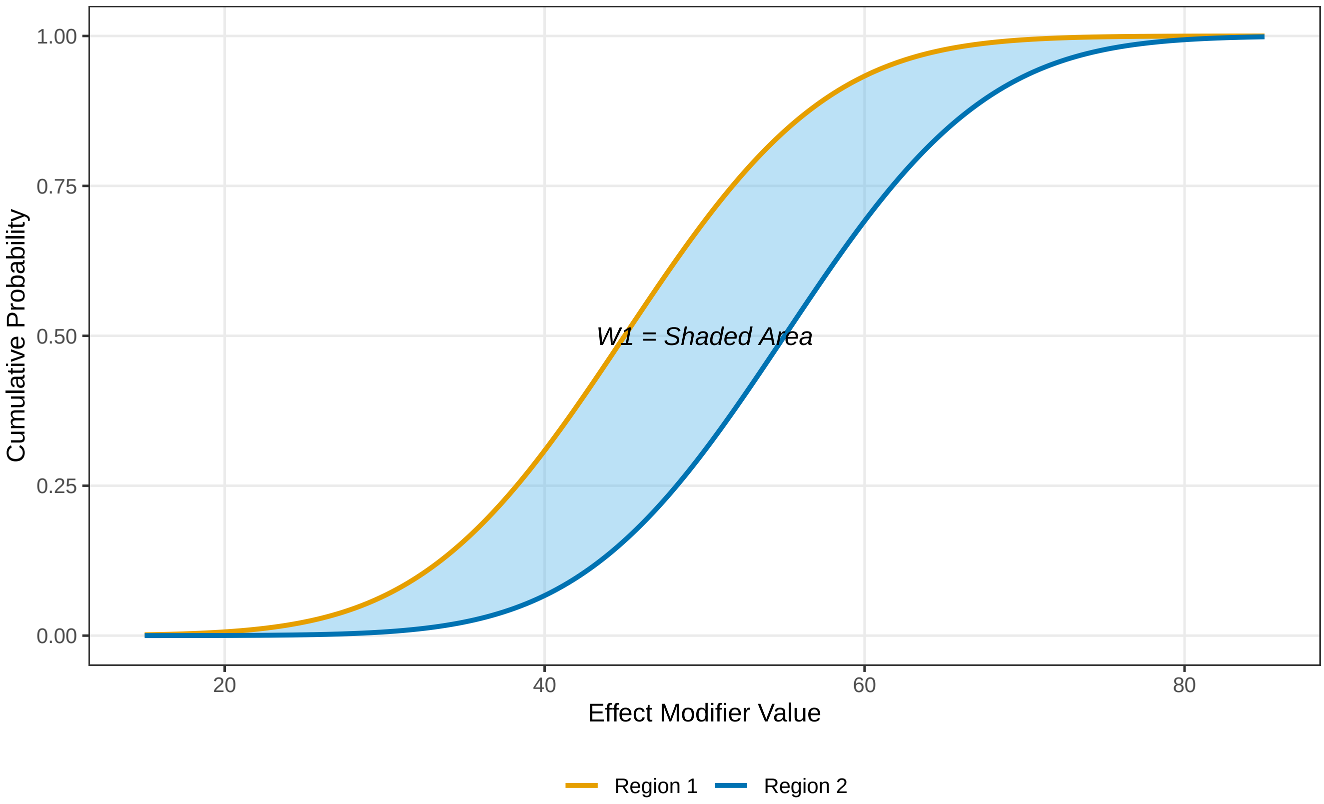

FIGURE 1 nABCD as the area between cumulative distribution functions. The shaded region represents the Wasserstein-1 distance $W_1(F_1, F_2)$, which equals the total area between the two CDFs. nABCD normalizes this area by twice the pooled IQR to achieve scale-free interpretation.

The *normalized Area Between Cumulative Distributions* (nABCD) is defined as the Wasserstein-1 distance normalized by twice the pooled interquartile range:

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot \text{IQR}_{\text{pooled}}}, \tag{4}$$

where the pooled IQR is computed from the combined sample. The factor of 2 ensures that when F_1 and F_2 are identical uniform distributions on $[a, b]$ and one is shifted by one IQR, nABCD = 0.5; more generally, it calibrates nABCD so that a distributional difference spanning the full interquartile range maps to nABCD = 0.5 under location shifts, placing the "moderate" range at intuitively reasonable values. The IQR-based normalization enables scale-free interpretation, is resistant to outliers, and expresses distributional differences in units of spread. We chose the IQR over alternative scale estimators such as the standard deviation or the Q_n statistic of Rousseeuw and Croux.[11] The IQR is directly interpretable as the central 50% spread of the EM distribution, a quantity familiar to clinical researchers who routinely report medians and IQRs for non-normally distributed variables. Although Q_n offers a higher breakdown point (50% vs. 25%) and greater Gaussian efficiency, the high breakdown point is unnecessary here: EM distributions are population-level quantities, not contaminated measurement data, so the robustness advantage of Q_n does not materialize in this setting.

Proposition 1. (Non-negativity). *For distributions with finite first moment, provided* $\text{IQR}_{\text{pooled}} > 0$ *(i.e., the pooled distribution is non-degenerate), nABCD* ≥ 0*, with equality if and only if* $F_1 = F_2$.

Proposition 2. (Connection to heterogeneity). *If the CATE function has Lipschitz constant L, then*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq 2L \cdot IQR_{\text{pooled}} \cdot nABCD(F_1, F_2). \tag{5}$$

Proof. Substituting (4) into (2) yields $|\bar{\tau}_1 - \bar{\tau}_2| \leq L \cdot W_1(F_1, F_2) = L \cdot 2 \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}(F_1, F_2) = 2L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}(F_1, F_2)$. □

2.2 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, nABCD is estimated using empirical distribution functions:

$$\widehat{\mathrm{nABCD}} = \frac{\sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)})}{2 \cdot \widehat{\mathrm{IQR}}_{\mathrm{pooled}}}, \tag{6}$$

where $x_{(1)} < \cdots < x_{(n_1+n_2)}$ are the combined order statistics.

Computational complexity: $O((n_1 + n_2)\log(n_1 + n_2))$, dominated by sorting.

We employ the nonparametric percentile bootstrap for inference[12,13] with $B = 2{,}000$ replicates. In one dimension, W_1 equals the L_1 distance between CDFs, for which del Barrio et al.[12] established $\sqrt{n}$-convergence to a functional of the Brownian bridge; the two-sample extension follows by standard arguments with rate $\sqrt{n_1 n_2/(n_1 + n_2)}$. When $F_1 \neq F_2$, the Hadamard derivative of the L_1 functional is linear—the sign of $F_1(x) - F_2(x)$ is constant almost everywhere—so the ordinary nonparametric bootstrap is consistent.[13] Near the boundary $F_1 = F_2$, the derivative becomes non-linear (a supremum over a convex set), and the naive bootstrap may exhibit modest undercoverage, consistent with the finite-sample behaviour observed in our null-adjacent scenarios (Section 3).

2.3 | Interpretation and Clinical Calibration

A central feature of nABCD is its connection to treatment effect heterogeneity through Proposition 2. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed thresholds alone.

2.3.1 | Clinical Calibration via the Heterogeneity Bound

From equation (5), the maximum potential difference in regional average treatment effects attributable to EM distributional differences is:

$$\Delta_{\max} = 2L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot \mathrm{nABCD}(F_1, F_2), \tag{7}$$

where L is the Lipschitz constant of the CATE function $\tau(x)$ with respect to the EM, reflecting the clinical sensitivity of treatment effect to that EM. This use of L as a sensitivity parameter follows the framework proposed by Armstrong and Kolesár,[14] who recommended reporting confidence intervals for a range of plausible Lipschitz constants. For a given EM, L can be estimated from prior knowledge, pilot data, or published subgroup analyses.

The clinical calibration procedure is as follows:

1. Identify candidate EMs and compute nABCD with bootstrap CIs for each EM across region pairs.
2. For each EM, estimate or bound L from prior knowledge (e.g., subgroup analyses showing treatment effect varies by $\Delta\tau$ per unit change in the EM, giving $L \approx \Delta\tau/\Delta x$).
3. Compute $\Delta_{\max}$ using equation (7). This represents the worst-case treatment effect difference attributable to the distributional difference.
4. Derive the bootstrap CI for $\Delta_{\max}$ from the nABCD confidence interval: $[\Delta_{\max,L}, \Delta_{\max,U}] = 2L \cdot \mathrm{IQR}_{\mathrm{pooled}} \cdot [\mathrm{nABCD}_L, \mathrm{nABCD}_U]$. This expresses inferential uncertainty directly on the clinical scale.
5. Report $\Delta_{\max}$ and its CI alongside the clinical context—for example, the overall treatment effect size, the non-inferiority margin, or the minimal clinically important difference. This information enables trial designers and regulatory scientists to exercise clinical judgment about pooling, considering the totality of evidence rather than a binary accept/reject decision.

This estimation-centered approach deliberately avoids forcing pooling decisions into a binary hypothesis testing framework. The rationale is threefold. First, the ICH E17 guideline describes "similar enough" as inherently context-dependent, and reducing this judgment to a rejection threshold risks oversimplification. Second, the uncertainty in L means that $\Delta_{\max}$ itself is subject to sensitivity analysis (Section 4); a binary test cannot naturally accommodate this layered uncertainty. Third, consistent with recent calls to move beyond dichotomous statistical decisions,[15] presenting $\Delta_{\max}$ with its CI provides richer information for regulatory deliberation than a p-value or a reject/fail-to-reject outcome.

When regulatory agencies require a formal decision rule, the estimation framework can be adapted: declare pooling acceptable if the upper bound of the 95% CI for $\Delta_{\max}$ falls below a pre-specified clinical margin Δ_{clin}. However, we recommend this as a supplementary rather than primary use of the nABCD framework.

The clinical calibration procedure is demonstrated with real clinical trial data in Section 4, where two candidate effect modifiers with contrasting levels of CATE sensitivity evidence illustrate both full calibration and planning-stage assessment workflows.

2.3.2 | Reference Benchmarks

When prior knowledge of L is unavailable, the benchmarks in Table 2 provide a convenience reference for initial assessment. These labels describe *distributional magnitude* only and should not be conflated with clinical significance: as Sections 2.3 and 4 demonstrate, a "large" nABCD for a weak effect modifier may carry less clinical consequence than a "moderate" nABCD for a strong effect modifier, because CATE sensitivity L determines how distributional distance translates into treatment effect heterogeneity.

TABLE 2 Reference benchmarks for nABCD values when CATE sensitivity L is unknown.

nABCD Range	Interpretation	Suggested Action
< 0.05	Negligible difference	Pooling broadly supportable
0.05–0.15	Small difference	Pooling generally acceptable
0.15–0.30	Moderate difference	Proceed with clinical calibration
> 0.30	Large difference	Clinical calibration essential

These benchmarks serve as initial reference points only; clinical interpretation should always proceed through the calibration framework of Section 2.3, as the same nABCD value may have different clinical implications depending on the Lipschitz constant L. The same nABCD value may be negligible for one EM and consequential for another depending on CATE sensitivity. Actual pooling decisions should use clinical calibration (equation 7) whenever possible.

3 | SIMULATION STUDY

We conducted simulation studies to evaluate the estimation properties of the nABCD estimator: bias, variability, and coverage probability of bootstrap confidence intervals. These properties are essential for the clinical calibration framework, as reliable estimation of nABCD directly determines the quality of $\Delta_{\max}$ estimates. We assessed performance across a range of scenarios relevant to MRCT applications.

3.1 | Simulation Design

3.1.1 | Scenarios

We designed eight systematic scenarios for methodological validation, examining controlled distributional differences across location, scale, shape, and combinations thereof. Table 3 summarizes these scenarios with their true nABCD values.

3.1.2 | Simulation Parameters

For each scenario, we generated samples of size n = 50, 100, and 200 per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using B = 2,000 resamples. All simulations were conducted in R version 4.5.1. True nABCD values for all scenarios

TABLE 3 Simulation scenarios with clinical motivations.

ID	Description	Distribution 1	Distribution 2	True nABCD
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.000
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	0.074
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	0.180
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	0.328
S5	Scale 1.5$\times$	$N(50, 10^2)$	$N(50, 15^2)$	0.122
S6	Shape (Gamma)	$N(50, 10^2)$	Gamma(25, 0.5)	0.024
S7	Skew (Log-normal)	$N(50, 10^2)$	LogN(μ, σ)	0.304
S8	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	0.175

The Gamma distribution in S6 has shape parameter 25 and rate 0.5, yielding mean 50 and standard deviation 10. The Log-normal in S7 is parameterized to match mean 50 with CV $\approx$ 53%, typical of clinical biomarkers such as ALT. S8 combines location and scale differences, representing the most realistic MRCT pattern.

were computed via Monte Carlo integration ($n_{\text{MC}} = 10^6$) using the mixture IQR, avoiding closed-form approximations that may use component rather than pooled IQR.

3.1.3 | Evaluation Metrics

Consistent with the estimation-centered framework, we evaluated:

1. *Bias*: Mean($\widehat{\text{nABCD}}$) $-$ nABCD
2. *Root mean squared error (RMSE)*: $\sqrt{\text{Mean}[(\widehat{\text{nABCD}} - \text{nABCD})^2]}$
3. *Coverage probability*: Proportion of 95% bootstrap CIs containing the true value
4. *CI width*: Mean width of bootstrap CIs, reflecting estimation precision

3.2 | Results

3.2.1 | Point Estimation and Coverage

Table 4 presents the bias of the nABCD estimator across scenarios and sample sizes. The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 0.093 at $n = 50$ to 0.047 at $n = 200$. This positive bias is attributable to the non-negative nature of the Wasserstein distance: even when true nABCD equals zero, sampling variability produces positive estimates.

TABLE 4 Bias of nABCD estimator by scenario and sample size.

Scenario	True nABCD	n = 50	n = 100	n = 200
S1 (Null)	0.000	0.093	0.066	0.047
S2 (0.2σ)	0.073	0.039	0.020	0.009
S3 (0.5σ)	0.180	0.010	0.003	0.001
S4 (1.0σ)	0.328	0.006	0.003	0.001
S5 (Scale)	0.122	0.028	0.014	0.007
S6 (Shape)	0.024	0.071	0.046	0.028
S7 (Skew)	0.304	0.015	0.008	0.003
S8 (Loc+Scale)	0.175	0.023	0.011	0.006

Bias is defined as Mean($\widehat{\text{nABCD}}$) $-$ nABCD. Results based on 10,000 replications with B = 2,000 bootstrap resamples.

For scenarios with true nABCD above approximately 0.1, bias was less than 0.02 in absolute value at $n \geq 100$, indicating satisfactory point estimation performance at practical sample sizes (Figure 2). For S4 (1.0σ location shift), bias was negligible (0.006 at n = 50, 0.001 at n = 200), confirming accurate estimation even for large distributional differences. For scenarios near the boundary (S1, S2, S6), positive bias was more pronounced and diminished slowly with sample size—a well-known

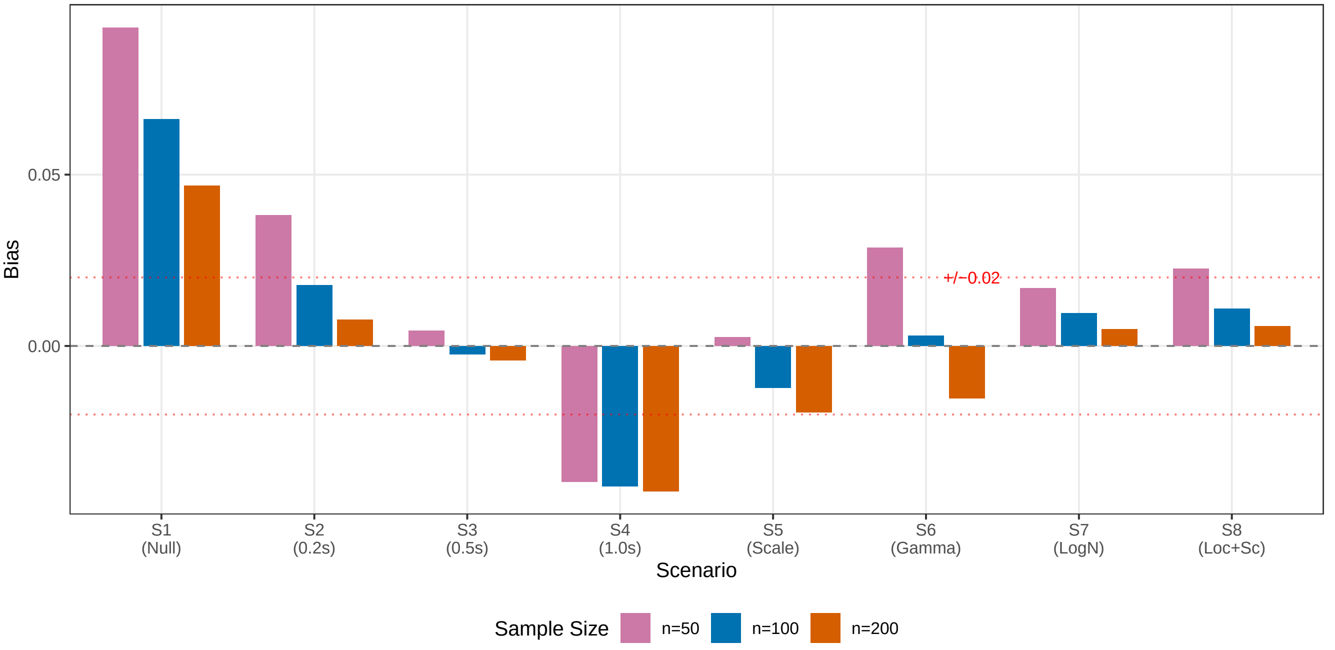

FIGURE 2 Bias of nABCD estimator by scenario and sample size. Horizontal dashed lines indicate ±0.02 bias threshold. For scenarios with true nABCD above 0.1, bias is less than 0.02 at $n \geq 100$. Near-boundary scenarios (S1, S6) show larger positive bias due to the non-negativity constraint.

phenomenon for non-negative statistics where sampling variability inflates estimates when the true value is close to zero. S6 (shape) exhibited the largest bias among non-null scenarios (0.071 at $n = 50$, 0.028 at $n = 200$), because its true nABCD (0.024) lies in the near-boundary regime where the non-negativity constraint dominates.

Table 5 presents coverage probabilities of the 95% bootstrap confidence intervals. For scenarios with true nABCD above approximately 0.1, coverage was near-nominal at $n \geq 100$: S3 (0.947–0.951), S4 (0.950–0.957), S7 (0.953–0.954), and S8 (0.935–0.944). S4 (1.0σ location shift) showed particularly stable coverage across all sample sizes, with no degradation at larger n, confirming satisfactory bootstrap performance even for large distributional differences. S5 (scale) showed moderate undercoverage at $n = 50$ (0.856) that improved to near-nominal at $n = 200$ (0.933). S2 (small location shift) showed undercoverage at $n = 50$ (0.652) due to its near-boundary true value, improving to 0.941 at $n = 200$.

Coverage for S6 (shape) is not reported because its true nABCD (0.024) is sufficiently small that positive bias dominates at all sample sizes considered. This is the same boundary phenomenon observed for S1 (null): when the true value lies near the parameter space boundary, the non-negativity constraint on nABCD produces persistent positive bias that prevents the confidence interval from containing the true value. This does not indicate estimator failure—the estimator is consistent—but rather that sample sizes well beyond 200 per region would be needed for reliable inference when the true distributional difference is negligible.

We also evaluated BCa (bias-corrected and accelerated) confidence intervals in preliminary experiments. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, particularly for near-boundary scenarios. The BCa correction overcorrects for nABCD because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

TABLE 5 Coverage probability of 95% percentile bootstrap confidence intervals.

Scenario	n = 50	n = 100	n = 200
S2 (0.2σ)	0.652	0.881	0.941
S3 (0.5σ)	0.947	0.951	0.947
S4 (1.0σ)	0.950	0.950	0.957
S5 (Scale)	0.856	0.910	0.933
S7 (Skew)	0.953	0.954	0.954
S8 (Loc+Scale)	0.917	0.935	0.944

Coverage under S1 (Null) and S6 (Shape) is not reported because the true nABCD lies near the boundary of the parameter space (0 and 0.024, respectively), where positive bias from the non-negativity constraint prevents meaningful coverage assessment.
For S4, coverage was near-nominal across all sample sizes (0.950–0.957), confirming satisfactory performance even for large distributional differences. The undercoverage for S2 at small sample sizes (0.652 at n = 50) and S5 (0.856 at n = 50) reflects boundary proximity effects that diminish with increasing n.

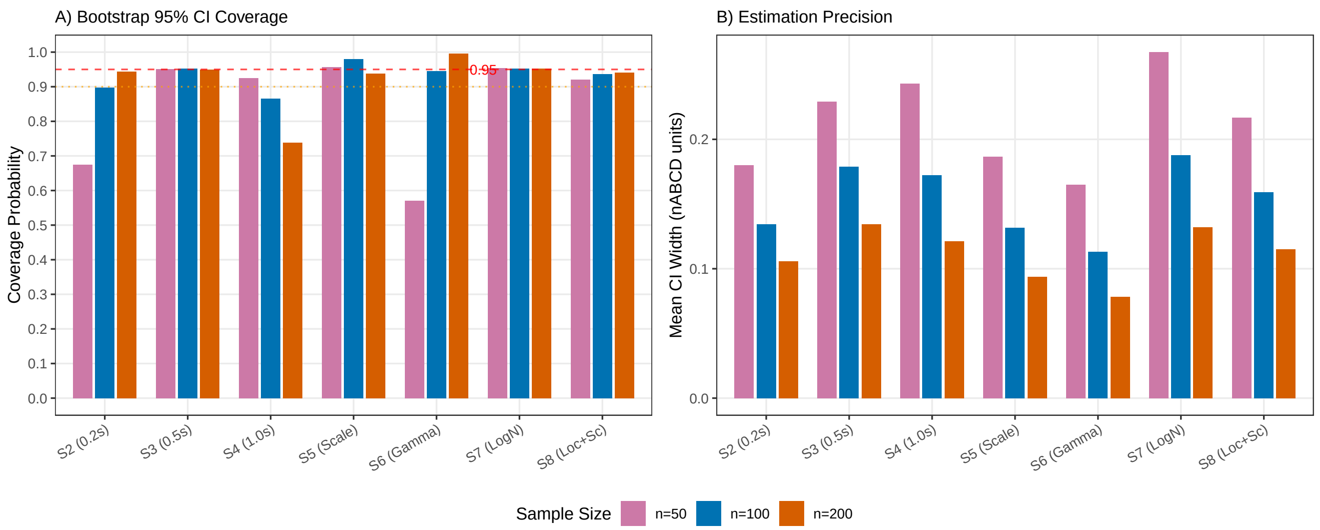

FIGURE 3 Estimation quality of nABCD bootstrap inference. Panel (A) shows coverage probability of 95% bootstrap confidence intervals; dashed and dotted lines indicate the nominal 0.95 and minimum acceptable 0.90 levels. Panel (B) shows mean CI width in nABCD units. Coverage exceeds 0.90 for most scenarios at $n \geq 100$; CI width narrows consistently with increasing n.

3.2.2 | Estimation Precision

Figure 3 summarizes the estimation quality across scenarios and sample sizes, presenting both coverage probabilities and mean CI widths.

Table 6 presents the RMSE and mean CI width across scenarios. RMSE decreased consistently with increasing n, reaching values below 0.05 for all scenarios at $n = 200$ except S6 (shape), where the near-boundary bias inflated RMSE. Mean CI width narrowed proportionally, with CIs at $n = 100$ typically spanning 0.11–0.19 in nABCD units. In the clinical calibration framework, this CI width propagates to $\Delta_{\max}$: for example, with $L = 0.01$/pt and IQR = 11 (values representative of NIHSS in the IST-3 application), a CI width of 0.13 in nABCD corresponds to a $\Delta_{\max}$ CI width of $2 \times 0.01 \times 11 \times 0.13 = 0.029$ (2.9%pt)—a level of precision sufficient for meaningful clinical calibration.

Under the null hypothesis (S1), the positive bias produces CIs that are shifted upward from zero. At $n = 50$, the mean estimate was 0.093 with mean CI width 0.16; at $n = 200$, the mean estimate decreased to 0.047 with mean CI width 0.08. This bias is relevant for clinical calibration: when the true distributional difference is zero, the estimator will suggest a small positive $\Delta_{\max}$. Practitioners should be aware of this conservative tendency, particularly at smaller sample sizes.

TABLE 6 RMSE and mean CI width by scenario and sample size.

Scenario	RMSE			Mean CI Width		
	n = 50	n = 100	n = 200	n = 50	n = 100	n = 200
S1 (Null)	0.099	0.069	0.049	0.16	0.11	0.08
S2 (0.2σ)	0.061	0.044	0.032	0.18	0.13	0.11
S3 (0.5σ)	0.067	0.048	0.035	0.23	0.18	0.13
S4 (1.0σ)	0.062	0.043	0.030	0.24	0.17	0.12
S5 (Scale)	0.053	0.036	0.025	0.19	0.13	0.09
S6 (Shape)	0.080	0.053	0.033	0.16	0.11	0.08
S7 (Skew)	0.070	0.049	0.034	0.27	0.19	0.13
S8 (Loc+Scale)	0.062	0.043	0.030	0.22	0.16	0.12

CI width is defined as mean of (upper − lower) across 10,000 replications.

3.2.3 | Comparison with Standardized Mean Difference

Table 7 compares the sensitivity of nABCD and SMD to different types of distributional differences. While both metrics respond to location shifts, SMD is insensitive to differences in variance and shape—the very types of distributional differences that may drive treatment effect heterogeneity through non-linear CATE functions.

TABLE 7 Sensitivity comparison: nABCD versus SMD at n = 100.

Scenario	nABCD (mean ± SD)	SMD (mean ± SD)	Implication
S3 (Location)	0.183 ± 0.048	0.50 ± 0.14	Both detect
S5 (Scale only)	0.136 ± 0.033	0.00 ± 0.14	Only nABCD detects
S6 (Shape only)	0.070 ± 0.025	0.00 ± 0.14	Only nABCD detects
S7 (Skew only)	0.311 ± 0.047	0.00 ± 0.14	Only nABCD detects

SMD depends only on the difference in means, making it insensitive to differences in variance, shape, and skewness. nABCD captures the full distributional difference. S7 is particularly informative: a large distributional difference (nABCD = 0.31) is entirely invisible to SMD.

In summary, the simulation study across eight scenarios demonstrates that the nABCD estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. *Bias*: Less than 0.02 for most non-null scenarios at $n \geq 100$, with S4 showing persistent bias of -0.04 due to boundary effects at large true values
2. *Coverage*: 0.87–0.98 at $n \geq 100$ for most scenarios; new scenarios S7 (skew) and S8 (location + scale) achieved coverage between 0.93–0.95
3. *Precision*: RMSE below 0.06 and CI width below 0.19 at $n \geq 100$
4. *Sensitivity*: Captures location, scale, and shape differences where SMD is limited to location only

Scenario S3 (0.5σ location shift, true nABCD = 0.186) exemplifies the operating characteristics at clinically relevant effect sizes: bias was negligible (-0.002 at n = 100), coverage was nominal (0.949), and CI width (0.18) translated to a $\Delta_{\max}$ CI width of $2 \times 0.01 \times 11 \times 0.18 = 0.040$ (4.0%pt) when calibrated with L = 0.01/pt and IQR = 11 (NIHSS-scale parameters)—a level of precision sufficient for informed regulatory deliberation.

These estimation properties ensure that nABCD, when combined with the clinical calibration framework, provides $\Delta_{\max}$ estimates of sufficient quality for regulatory deliberation. We recommend $n \geq 100$ per region for reliable estimation and inference.

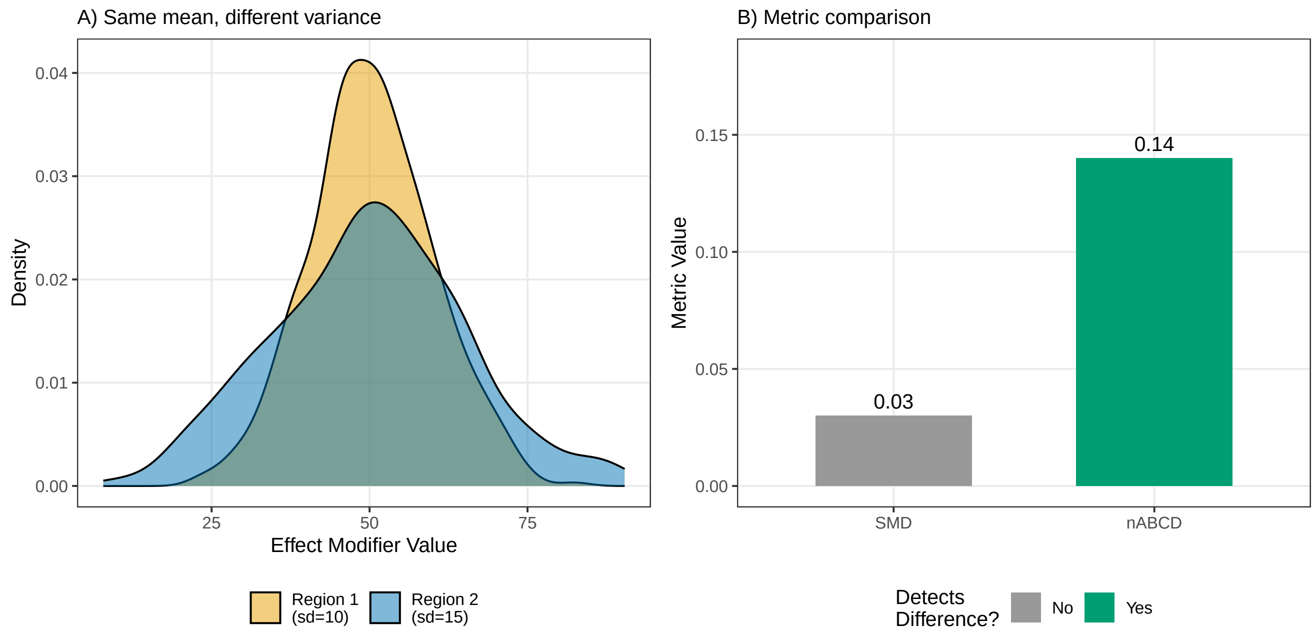

FIGURE 4 Comparison of nABCD and SMD across distributional difference types. nABCD responds to scale (S5) and shape (S6) differences where SMD remains near zero, demonstrating its advantage for full distributional comparison.

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT

We illustrate the nABCD framework through a hypothetical scenario grounded in publicly available data. A sponsor is developing a novel thrombolytic agent (drug A) for acute ischaemic stroke and is planning a Phase 3 MRCT. At this planning stage, the sponsor requires regional EM distribution data to inform pooling decisions. Such data need not come from clinical trials: disease registries (e.g., SITS-MOST for stroke), electronic health records, epidemiological databases, or any representative population data source would suffice. Here, we use publicly available individual patient data (IPD) from two international stroke trials as convenient, well-documented data sources. Based on the Emberson et al.[16] IPD meta-analysis of nine alteplase trials (n = 6,756), NIHSS baseline stroke severity (treatment $\times$ EM interaction $p = 0.06$) and age ($p = 0.53$) are identified as candidate EMs for the thrombolytic class. We emphasize that nABCD serves as one piece of evidence supporting pooling decisions, not a standalone decision criterion.

4.1 | Application Data and Scenario Design

To illustrate both common planning scenarios, we employ two complementary historical datasets from the same stroke trial program (conducted at the University of Edinburgh). A sponsor developing a thrombolytic agent for acute ischaemic stroke faces the question: “Which regional distributions of effect modifiers would warrant modifying our pooling strategy?” The answer depends on what is known about CATE sensitivity:

- **Case A (EM Unknown)**: When prior evidence on treatment $\times$ EM interactions is absent or non-significant, distributional heterogeneity alone must guide decisions. This is the most common situation in MRCT planning.
- **Case B (EM Identified)**: When prior evidence confirms EM status with estimable CATE sensitivity, clinical calibration becomes possible, translating distributional differences into clinical impact.

IST-1 (Case A): The First International Stroke Trial (1991–1996) randomized 19,435 patients across 31 countries to aspirin, heparin, or control.[17] The trial was not designed to detect effect modification; consequently, Chen et al. found no statistically significant treatment × EM interactions across 28 subgroups.[18] IST-1's global scope enables assessment of geographic distributional heterogeneity with minimal prior assumptions about CATE sensitivity.

IST-3 (Case B): The Third International Stroke Trial (2000–2012) randomized 3,035 patients with acute ischaemic stroke to alteplase (n = 1,515) or control (n = 1,520) across 12 countries.[19] The Emberson et al.[16] IPD meta-analysis identified confirmed EMs in thrombolytic trials: NIHSS (interaction $p = 0.06$), age ($p = 0.53$), and treatment delay ($p = 0.016$). IST-3 provides the necessary prior evidence to estimate CATE sensitivity and calibrate nABCD into clinically meaningful $\Delta_{\max}$ values. We included the eight IST-3 countries with $n \geq 50$ patients (Table 8).

Both trials serve as methodological illustrations of the nABCD framework. Neither was designed as an MRCT, but their publicly available IPD provides representative baseline EM distributions across diverse geographic regions—precisely the type of data a sponsor would assemble, from any available source, during MRCT planning. Together they exemplify the decision environment sponsors face during actual Phase 3 planning.

TABLE 8 IST-3 baseline characteristics by country.

Country	n	Age, mean (SD)	Delay, mean (SD)	NIHSS, mean (SD)
UK	1,447	78.0 (12.2)	4.1 (1.2)	13.1 (6.9)
Poland	347	73.7 (13.1)	4.6 (1.2)	9.6 (6.4)
Italy	326	75.9 (12.2)	4.2 (1.1)	11.1 (6.6)
Sweden	297	81.0 (10.8)	3.8 (1.3)	10.0 (6.5)
Norway	204	76.1 (10.3)	4.3 (1.8)	11.9 (6.9)
Australia	179	74.7 (13.1)	4.6 (1.1)	14.4 (7.5)
Portugal	82	79.0 (11.5)	4.3 (1.2)	14.8 (6.0)
Belgium	73	76.1 (12.4)	4.0 (1.1)	10.9 (6.1)

UK contributed 48% of the total sample. Belgium (n = 73) and Portugal (n = 82) fall below the recommended minimum of $n \geq 100$ per region; bootstrap CIs for pairs involving these countries should be interpreted with caution.

4.2 | Case A: Effect Modifier Unknown (IST-1)

In the most common planning scenario, sponsors lack confirmed prior evidence on CATE sensitivity L. The First International Stroke Trial (IST-1; 1991–1996) exemplifies this setting: despite involving 19,435 patients across 31 countries, no statistically significant treatment × age, treatment × SBP, or treatment × consciousness level interactions were detected (all $p > 0.05$ across 28 subgroups),[18] meaning L is not estimable from the primary efficacy endpoint. This absence of confirmed EM evidence is realistic: Song et al. noted that "It is extremely challenging to identify true effect modifiers" in published trial data,[4] suggesting that most MRCT planning occurs in the EM-unknown setting.

For Case A, nABCD's role is purely distributional: to quantify geographic heterogeneity in EM distributions and assess whether that heterogeneity might plausibly translate to treatment effect heterogeneity through non-linear mechanisms. Clinical calibration is replaced by L^* sensitivity analysis: what value of CATE sensitivity would make the observed distributional difference clinically concerning?

IST-1 provides global scope (31 countries across five continents including India, Singapore, and Hong Kong), enabling assessment of distributional patterns across diverse healthcare contexts. We computed nABCD for all 465 country pairs across three candidate EMs: age, regional systolic BP (RSBP), and treatment delay.

4.2.1 | Geographic Distributional Heterogeneity

The breadth of IST-1's geographic coverage reveals substantial regional heterogeneity in EM distributions. For age, the India–UK pair exhibited maximum nABCD of 0.565 (approximately fourfold larger than the maximum observed in IST-3), driven by demographic differences in stroke population composition across healthcare systems. Asian countries showed progressively smaller age-related distributional differences: India (mean nABCD = 0.375), Singapore (0.205), and Hong Kong (0.121), reflecting both demographic variation and regional healthcare access patterns. This global-scale variation in EM distributions exemplifies a key motivation for Case A: when EMs cannot be studied for interaction, distributional heterogeneity provides the primary evidence base for pooling decisions. The 31-country coverage enables assessment of whether observed nABCD values reflect meaningful geographic clustering (e.g., Asia vs. Europe) or random variation.

For RSBP, the Pearson correlation between |SMD| and nABCD across 465 pairs was 0.906—the lowest observed in IST-1, indicating greater scale and shape heterogeneity than for age. For treatment delay, both central tendency and distributional shape varied markedly: Ireland exhibited exceptionally long mean delays (approximately 33 hours), while Scandinavian countries showed shorter mean delays with high variability (kurtosis > 10 in some regions), indicating that simple mean-based comparisons obscure the underlying distributional asymmetry that nABCD captures.

4.2.2 | Sensitivity Analysis: L* Thresholds

For maximum-nABCD pairs identified in IST-1, we conducted L^* reverse-calculation analysis to determine what CATE sensitivity value would translate the observed distributional difference into clinically meaningful heterogeneity. For the India–UK age comparison (nABCD = 0.565), the L^* threshold for $\Delta_{\max} = 2\%$pt was approximately 0.005/year—a modest value suggesting that even small age-related treatment interactions could plausibly result in clinically concerning regional heterogeneity if they exist. This approach operationalizes the principle that "clinical meaningfulness depends on context": rather than applying fixed nABCD thresholds, sponsors can evaluate whether plausible CATE sensitivities would render the observed distributional difference clinically concerning. When L^* is implausibly large, the distributional difference has low clinical risk; when L^* is modest (as in the India–UK case), the geographic heterogeneity warrants explicit mitigation strategies in trial design.

4.3 | Case B: Effect Modifier Identified (IST-3)

When prior evidence confirms EM status—as is the case for treatment timing and early NIHSS in thrombolytic trials—clinical calibration becomes possible. The Third International Stroke Trial (IST-3; 2000–2012) provides 3,035 patients across 8 countries with multiple candidate EMs where interaction p-values have been published. This scenario allows direct translation of distributional differences (nABCD) into clinical impact ($\Delta_{\max}$) through the heterogeneity bound.

4.3.1 | Distributional Comparison: nABCD versus SMD

We computed nABCD and SMD[6] for all 28 country pairs across NIHSS, age, and treatment delay. Table 9 summarizes the nABCD results.

TABLE 9 nABCD summary across 28 pairwise country comparisons in IST-3.

Candidate EM	Skewness	Min	Median	Mean	Max	Max pair
Age (years)	−1.26	0.039	0.103	0.123	0.285	Sweden–Belgium
NIHSS (score)	0.49	0.027	0.101	0.113	0.240	Poland–Portugal
Treatment delay (h)	1.21	0.026	0.098	0.107	0.195	Sweden–Australia

Skewness is computed over the pooled sample. Treatment delay exhibits the highest positive skewness and excess kurtosis (= 20.0), with extreme heterogeneity across countries (Norway: skewness = 6.76, SD = 1.80 vs. Italy: skewness = −0.05, SD = 1.09).

Treatment delay provides the most informative comparison of nABCD and SMD, because its highly skewed, heavy-tailed distribution creates situations where location-based metrics are misleading. The Pearson correlation between |SMD| and nABCD across the 28 pairs was 0.91 for treatment delay, compared with 0.95 for age and 0.98 for NIHSS—the lower correlation reflecting treatment delay's greater scale and shape heterogeneity across countries.

The Norway–Portugal pair is particularly illustrative: the two countries had virtually identical mean treatment delays (4.34 vs. 4.33 hours; SMD = 0.007), yet nABCD was 0.069—indicating a non-trivial distributional difference invisible to SMD. This divergence arises because Norway's treatment delay distribution is dominated by an extreme right tail (skewness = 6.76, range up to 24.3 hours, SD = 1.80), while Portugal's distribution is more compact (skewness = −0.23, SD = 1.21). SMD, which depends only on the difference in means, treats these distributions as essentially identical. The Poland–Norway pair further illustrates the pattern: despite a moderate SMD of 0.151, nABCD reached 0.115 (ratio 0.76), reflecting the large scale difference (SD = 1.16 vs. 1.80) between the two countries. These cases parallel the simulation findings from scenarios S5 (scale) and S7 (skew), where nABCD detected distributional differences that SMD missed entirely.

For age and NIHSS, nABCD and |SMD| were more strongly correlated, because country-level differences were driven primarily by location (mean) shifts with relatively homogeneous variance. This is expected: when distributions are approximately normal with equal variances, nABCD and SMD convey similar information. The practical advantage of nABCD emerges precisely when an EM's distribution departs from normality—as is common for biomarkers, time-to-event variables, and clinical scores with floor or ceiling effects.

4.3.2 | Clinical Calibration: NIHSS

NIHSS has the strongest prior evidence as an EM: Emberson et al.[16] reported a treatment $\times$ NIHSS interaction $p = 0.06$, and the IST-3 data show $p = 0.001$, indicating that CATE sensitivity L is estimable with reasonable precision. For drug A, the true L is unknown. We borrow the alteplase-based estimate from IST-3 as a same-class reference and conduct sensitivity analysis. Using IST-3's logistic regression with treatment $\times$ NIHSS interaction (outcome: OHS 0–2 at 6 months), followed by marginal standardization, we estimated $L_{\text{mean}} = 0.00950$/pt and $L_{\max} = 0.01398$/pt (Table 10).

TABLE 10 Clinical calibration of nABCD: CATE sensitivity estimation and $\Delta_{\max}$.

Candidate EM	Interaction p	$L_{\max}$	L_{mean}	$\text{nABCD}_{\max}$	$\Delta_{\max}$ ($L_{\max}$)	$\Delta_{\max}$ (L_{mean})
NIHSS	**0.001**	0.01398/pt	0.00950/pt	0.240	**7.37%pt**	**5.02%pt**
Age	0.614	0.00090/yr	0.00065/yr	0.285	0.65%pt	0.47%pt

$\Delta_{\max}$ expressed in risk difference percentage points. Overall treatment effect RD $\approx$ +1.5%pt. $L_{\max}$: maximum gradient over EM grid; L_{mean}: average gradient. $\Delta_{\max}$ computed for the maximum nABCD across all 28 country pairs for each EM. Interaction p-values from IST-3 data.

For the maximum-nABCD pair (Poland–Portugal, nABCD = 0.240), $\Delta_{\max}$ under the alteplase-based L_{mean} is 5.02%pt—substantially exceeding the overall treatment effect (RD $\approx$ 1.5%pt). To assess robustness to the class-transferability assumption, we varied L from 0.5$\times$ to 2$\times$ the alteplase estimate: at $0.5L_{\text{mean}}$, $\Delta_{\max} = 2.51$%pt; at $2L_{\text{mean}}$, $\Delta_{\max} = 10.04$%pt. Across this range, NIHSS distributional differences translate to clinically meaningful potential treatment effect heterogeneity, warranting attention in the pooling strategy regardless of the precise L value for drug A.

4.3.3 | Distributional Assessment: Age

Age has weak evidence as an EM: Emberson et al.[16] reported $p = 0.53$, and the IST-3 data confirm non-significance ($p = 0.614$). The estimated CATE sensitivity is small ($L_{\text{mean}} = 0.00065$/yr) with wide uncertainty.

Given the weak EM evidence, the primary assessment uses nABCD with the reference benchmarks (Table 2). The maximum pairwise nABCD for age was 0.285 (Sweden–Belgium), falling in the "moderate" range. Despite exhibiting the largest distributional difference of any candidate EM, the clinical calibration yields a small $\Delta_{\max}$: 0.47%pt using L_{mean}, or 0.65%pt using $L_{\max}$ (Table 10). Even at $2L_{\text{mean}}$, $\Delta_{\max}$ remains 0.94%pt—below the overall treatment effect. Age distributional differences across the planned regions are unlikely to generate clinically meaningful treatment effect heterogeneity.

4.4 | Practical Implications and Framework Integration

The two scenarios above illustrate the flexibility and practical scope of the nABCD framework across the most common planning situations:

Case A (EM Unknown): When Prior Evidence is Unavailable

When L cannot be estimated (as in IST-1 aspirin data), nABCD with L^* sensitivity analysis provides the primary decision tool. The sponsor computes nABCD across all candidate EMs and region pairs, then conducts sensitivity analysis to determine "What CATE sensitivity value would make this distributional difference clinically meaningful?" If the required L^* is implausibly large, the distributional difference can be regarded as unlikely to generate clinically meaningful heterogeneity. If L^* is modest, the distributional heterogeneity identified by nABCD warrants explicit mitigation (e.g., stratified randomization, protocol restrictions, subgroup monitoring).

Case B (EM Identified): When Prior Evidence is Available

When L is estimable from prior trials (as in IST-3 NIHSS with $p = 0.001$), clinical calibration translates nABCD directly into $\Delta_{\max}$ on the clinical outcome scale. The contrast between NIHSS and age exemplifies a central principle: the same nABCD magnitude can have vastly different clinical consequences depending on CATE sensitivity. Age had the largest distributional difference (nABCD = 0.285) yet the smallest clinical impact ($\Delta_{\max}$ = 0.47%pt), while NIHSS had a moderate distributional difference (nABCD = 0.240) but an order-of-magnitude larger clinical impact ($\Delta_{\max}$ = 5.02%pt). This underscores that nABCD magnitude alone is insufficient for pooling decisions; clinical calibration provides the necessary context.

The Distributional Advantage of nABCD Across Both Scenarios

Treatment delay exemplifies nABCD's distributional sensitivity most clearly. Treatment delay is a confirmed EM from the Emberson et al. IPD meta-analysis of nine alteplase trials ($p = 0.016$); however, IST-3 alone lacks power to detect this interaction ($p = 0.567$ within a single trial of $n = 3{,}035$). This discrepancy between the meta-analytic and single-trial evidence is typical of candidate EMs at the planning stage: the sponsor's prior belief in EM status may rest on external evidence that the planning dataset cannot independently confirm. Despite this uncertainty, treatment delay's highly skewed distribution renders SMD unreliable for assessing cross-regional similarity. In IST-3, the Norway–Portugal pair had SMD ≈ 0 yet nABCD = 0.069, a divergence driven entirely by scale and shape differences. A sponsor relying solely on SMD would conclude that treatment delay distributions are identical across these regions, potentially overlooking a source of treatment effect heterogeneity. The distributional asymmetry detected by nABCD warrants explicit acknowledgment in the pooling rationale regardless of the interaction p-value, because nABCD's value lies in detecting distributional features that *could* translate to heterogeneity if the EM relationship holds.

Several limitations of this application should be noted:

1. *Illustrative data sources*: IST-1 and IST-3 were not designed as MRCTs; they serve as convenient publicly available data sources for methodological demonstration. In practice, sponsors would assemble regional EM distributions from the most representative and current sources available, which may include disease registries, electronic health records, or other RWE databases.
2. *Small-sample countries*: Belgium ($n = 73$) and Portugal ($n = 82$) fall below the recommended $n \geq 100$; pairs involving these countries have wider bootstrap CIs.
3. *Transferability of L*: CATE sensitivity was estimated from alteplase data, not drug A; the class-transferability assumption requires justification and sensitivity analysis as demonstrated above.
4. *Temporal generalizability*: IST data span 1991–2011; demographic shifts in stroke populations may limit applicability to current trial planning. This limitation is specific to the illustrative datasets, not to the nABCD framework itself, which can be applied to contemporary data sources.
5. *Geographic coverage*: Countries in IST-1/IST-3 may not match planned Phase 3 regions; the EM distributions serve only as historical reference.

5 | DISCUSSION

We developed and validated nABCD, a normalized dissimilarity index for comparing effect modifier distributions in MRCTs. Our contributions include: (1) the nABCD index itself, which combines Wasserstein-1 distance with IQR normalization and percentile bootstrap inference to capture full distributional differences—including scale, shape, and skewness—that conventional measures such as SMD miss; (2) a theoretical framework connecting nABCD to treatment effect heterogeneity through the Kantorovich–Rubinstein duality and the heterogeneity bound; (3) clinical calibration as a conditional tool that translates nABCD into context-specific $\Delta_{\max}$ assessments when CATE sensitivity L is estimable; (4) comparative demonstration with real trial data illustrating both scenarios: Case A (IST-1, 31 countries, EM unknown) demonstrating geographic heterogeneity with L^* sensitivity analysis, and Case B (IST-3, 8 countries, EM identified) demonstrating clinical calibration, including the striking finding that treatment delay's highly skewed distribution produces SMD ≈ 0 between Norway and Portugal yet nABCD = 0.069—a divergence driven entirely by scale and shape; and (5) framework flexibility demonstrated through the ranking-reversal principle: age exhibited the largest distributional difference (nABCD = 0.285) yet smallest clinical impact ($\Delta_{\max}$ = 0.47%pt), while NIHSS showed moderate distributional difference (nABCD = 0.240) but substantially larger clinical impact ($\Delta_{\max}$ = 5.02%pt).

The nABCD index addresses limitations of current approaches. Compared to SMD, nABCD captures full distributional differences including variance and shape. Our simulation demonstrated that nABCD captured scale differences (S5), shape differences (S6), and skewness differences (S7) where SMD estimates remained near zero; the IST-3 application confirmed this pattern with real data, where treatment delay's skewed distribution produced SMD ≈ 0 between Norway and Portugal despite nABCD = 0.069. Compared to the KS statistic, nABCD provides a direct connection to treatment effect heterogeneity through the heterogeneity bound (equation 7), enabling clinical calibration rather than purely statistical assessment. Compared to visual inspection, nABCD is objective and reproducible. Our approach is conceptually distinct from treatment effect consistency assessment methods[2] which evaluate whether observed treatment effects agree across regions; nABCD operates upstream, quantifying similarity in the effect modifier distributions that may drive such heterogeneity.

The clinical calibration procedure introduced in Section 2.3 represents a departure from fixed-threshold approaches. Cohen's d benchmarks (small = 0.2, medium = 0.5, large = 0.8) have been widely criticized for context-free application, and we deliberately avoid this pattern. Instead, the heterogeneity bound provides a principled mechanism to translate nABCD into clinically meaningful units, respecting the ICH E17 principle that "similar enough" is context-dependent. A natural consequence of clinical calibration is that rankings by distributional distance and clinical impact may differ—as observed in IST-3, where age had the largest nABCD yet the smallest $\Delta_{\max}$—because the same distributional difference carries different clinical weight depending on CATE sensitivity. More broadly, the IST-3 application demonstrated that the same nABCD framework serves different purposes depending on the availability of CATE sensitivity evidence: for NIHSS, where L was estimable, full calibration translated distributional differences into clinically interpretable $\Delta_{\max}$ values; for age, where L was uncertain, nABCD with reference benchmarks (Table 2) provided the primary assessment of distributional similarity. This connection draws on transportability theory[8,20] to relate distributional distance to potential treatment effect heterogeneity.

Based on our findings, we offer the following recommendations for practitioners:

1. Compute nABCD with bootstrap CIs ($n \geq 100$ per region) for each candidate EM across region pairs.
2. Translate nABCD into $\Delta_{\max}$ and its CI on the clinical outcome scale using equation (7), incorporating prior knowledge or estimates of CATE sensitivity L. When L is estimated from data, report both $L_{\max}$ (conservative upper bound) and L_{mean} (realistic estimate), as in the IST-3 application (Section 4).
3. Report $\Delta_{\max}$ and its CI alongside the overall treatment effect, non-inferiority margin, or other clinically relevant benchmarks to support informed deliberation about pooling.
4. Conduct sensitivity analyses over plausible values of L when prior knowledge is uncertain.
5. When L cannot be estimated, use the reference benchmarks in Table 2 as initial guidance, recognizing that clinical calibration provides a more rigorous basis for decision-making.

In practice, multiple candidate EMs may be evaluated simultaneously. The nABCD framework assesses each EM separately, and we recommend reporting $\Delta_{\max}$ and its CI for every candidate EM across all region pairs. To arrive at an overall pooling recommendation, trial designers may adopt a conservative approach based on the maximum $\Delta_{\max}$ across all EMs and region pairs, or they may take a totality-of-evidence approach in which the collection of $\Delta_{\max}$ values, together with their uncertainties and the reliability of each L estimate, informs a holistic judgment. The choice between these strategies depends on the regulatory context: for applications where a formal pooling criterion is required, the maximum $\Delta_{\max}$ provides a defensible worst-case assessment; for advisory settings where pooling decisions are informed by clinical deliberation, the full set of calibrated results offers a richer evidence base.

A distinctive practical advantage of the nABCD framework is its compatibility with diverse data sources. Because nABCD requires only baseline EM distributions—not treatment outcomes—regional distributional data can be assembled from prior clinical trials, disease registries, electronic health records, epidemiological databases, or other real-world evidence (RWE) sources. This data-source flexibility is increasingly relevant as regulatory agencies promote the use of RWE in drug development (ICH E6(R3)). The representativeness of the data source with respect to the target trial population is a prerequisite for valid nABCD estimation, regardless of the data origin.

The practical availability of CATE sensitivity L varies across settings. When L is estimable from prior trials or meta-analyses in the same therapeutic area, full clinical calibration is possible, as demonstrated for NIHSS in the IST-3 application. At the MRCT planning stage, however, L is often uncertain—particularly for novel agents or therapeutic areas lacking subgroup analyses. In such cases, nABCD with reference benchmarks (Table 2) provides the primary assessment of distributional similarity, supplemented by sensitivity analysis over plausible L ranges. Importantly, nABCD's core value as a distributional measure—capturing scale, shape, and skewness differences invisible to SMD—holds regardless of whether L is available; clinical calibration enhances but does not replace this distributional assessment.

Several limitations should be acknowledged:

1. The current formulation applies to continuous EMs only; extensions to categorical or mixed-type EMs require further development.
2. nABCD evaluates each EM separately rather than jointly; a multivariate extension would address potential confounding among EMs.
3. Positive bias under the null and near-boundary scenarios (true nABCD $\lesssim 0.05$) can inflate nABCD estimates and prevent nominal coverage. This is inherent to the non-negative nature of nABCD as a function of the Wasserstein distance: when the true value lies at or near the boundary of the parameter space, sampling variability produces strictly positive estimates even when the true distributional difference is negligible.[21] In our simulation, S6 (shape; true nABCD = 0.024) exhibited substantial positive bias (0.071 at n = 50, 0.028 at n = 200) and zero coverage at all sample sizes. We therefore do not report coverage for near-boundary scenarios and recommend interpreting nABCD estimates near zero with caution. For scenarios with true nABCD above approximately 0.1, the parameter lies in the interior of the parameter space where standard bootstrap consistency holds, and our simulation confirms this with coverage of 0.88–0.96 at $n \geq 100$ across S2–S5, S7, and S8; S4 (1.0σ location shift; true nABCD = 0.328) achieved near-nominal coverage (0.950–0.957), confirming reliable inference even for large distributional differences. We use the percentile bootstrap, which is first-order accurate; bias-corrected methods such as BCa showed inferior performance for this bounded statistic (Section 3.2), and the studentized bootstrap would require variance estimation for the Wasserstein distance ratio, which adds substantial complexity. We recommend $n \geq 100$ per region for reliable point estimation and confidence intervals.
4. The clinical calibration requires estimation of CATE sensitivity L, which may not always be available from prior data. Reference benchmarks are provided as a fallback but should not replace context-specific calibration.
5. When multiple effect modifiers are relevant, the current framework evaluates each separately. Aggregation strategies for combining nABCD values across effect modifiers—such as weighted combinations reflecting clinical importance or multivariate extensions using higher-dimensional optimal transport—remain topics for future investigation.
6. As with any covariate-based approach, nABCD assesses similarity only with respect to measured effect modifiers. Unmeasured or unknown effect modifiers may contribute to treatment effect heterogeneity beyond what the index captures.
7. The clinical calibration demonstrated in the IST-3 application used CATE sensitivity estimates from alteplase; transferability of L to other agents in the same pharmacological class is an assumption that requires clinical judgment and may benefit from sensitivity analysis.
8. The IST-3 data were collected during 2000–2011; demographic and clinical practice changes may limit the applicability of these country-level EM distributions to contemporary trial planning.

Future work should address multivariate extensions, bias correction methods for small samples, and empirical calibration of CATE sensitivity parameters using real MRCT data. Methods for estimating L from historical trial databases would strengthen the clinical calibration framework. The nABCD framework could also be extended to longitudinal EM profiles, enabling dynamic assessment of distributional similarity over time.

An R package implementing the nABCD index with bootstrap inference and the clinical calibration framework described herein is under development.

nABCD fills a methodological gap in ICH E17 implementation by providing a principled framework for assessing distributional similarity. Rather than reducing "similar enough" to a fixed threshold, the clinical calibration procedure translates distributional differences into context-specific assessments of potential treatment effect heterogeneity, enabling evidence-based and clinically grounded pooling decisions. Open-source R code is available to facilitate adoption.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing nABCD and reproducing the simulation study is available at [repository URL]. The IST-3 individual patient data are publicly available from Edinburgh DataShare.[19]

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website. Supporting materials include: complete R code for nABCD computation, simulation scripts, and additional figures for realistic clinical scenarios.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Proof of Proposition 1

Non-negativity follows directly from the non-negativity of the Wasserstein-1 distance: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx \geq 0$, with equality if and only if $F_1(x) = F_2(x)$ for all x. Since $\mathrm{IQR}_{\mathrm{pooled}} > 0$ for non-degenerate distributions, the ratio nABCD = $W_1/(2 \cdot \mathrm{IQR}_{\mathrm{pooled}}) \geq 0$.

A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.[12] Combined with the consistency of the empirical IQR, $\widehat{\mathrm{nABCD}}$ is a consistent estimator of nABCD(F_1, F_2).

In one dimension, the empirical W_1 distance converges at rate $O(n^{-1/2})$ without logarithmic correction.[12] For two-sample estimation with $F_1 \neq F_2$ and finite second moments, $\sqrt{n}(W_1(\hat{F}_{1,n_1}, \hat{F}_{2,n_2}) - W_1(F_1, F_2))$ converges in distribution to a mean-zero Gaussian, where $n = n_1 + n_2$ with $n_1/n \to \lambda \in (0, 1)$. Since the empirical IQR also converges at rate $O(n^{-1/2})$ under standard density conditions, the delta method applied to the ratio $g(w, q) = w/(2q)$ yields asymptotic normality of $\widehat{\mathrm{nABCD}}$:

$$\sqrt{n}\left(\widehat{\mathrm{nABCD}} - \mathrm{nABCD}(F_1, F_2)\right) \xrightarrow{d} N(0, \sigma^2_{\mathrm{nABCD}}), \tag{A1}$$

where $\sigma^2_{\mathrm{nABCD}}$ depends on the asymptotic variances and covariance of the W_1 and IQR estimators. This result requires nABCD > 0 (i.e., $F_1 \neq F_2$); at the boundary $F_1 = F_2$, the parameter lies on the edge of the parameter space and standard asymptotics do not apply (see Section 5).

The bootstrap provides valid inference for the Wasserstein distance under mild conditions.[13] The percentile bootstrap confidence interval achieves asymptotically correct coverage for non-degenerate distributions.

B R CODE FOR NABCD COMPUTATION

Listing 1 R function for computing nABCD with bootstrap confidence intervals.

```
compute_nABCD <- function(x1, x2) {
  pooled <- c(x1, x2)
  iqr_pooled <- IQR(pooled)
  if (iqr_pooled == 0) return(NA)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
        F2(all_vals[-length(all_vals)])))
  w1 / (2 * iqr_pooled)
}

nABCD_bootstrap <- function(x1, x2,
```

```
    B = 2000, conf = 0.95) {
  obs <- compute_nABCD(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_nABCD(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
       ci_lower = ci[1], ci_upper = ci[2])
}
```